AGENT DESIGN STUDIO

Daily UTC delivery lateness CSV report

PASS | DESIGN ONLY

A reviewed system-design specification,not an implemented system or executed business test. Scores below are supplied design judgements.

Review: self-review / round 3 | 81 canonical cards across ten stages.

This report prioritizes design calls,contracts and review evidence. Full fixtures and exact canonical wording remain in design.json; eval-set.json contains the complete machine-readable evaluation set.

Design dimension	Score	Design dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	4/4
D3	3/4	D9	3/4
D4	3/4	D10	3/4
D5	3/4	D11	3/4
D6	3/4	D12	3/4

1 Context

Original brief

Design a small daily report that joins two CSV files by order ID. One has promised-delivery timestamps and the other actual-delivery timestamps, both ISO UTC. Flag delivered orders more than 24 hours late and export a CSV for the operations team. No emails, remote updates, forecasting or conversational interface.

CTX1 - Business decision and actors

Business decision and actors: Operations uses the exported daily CSV to identify which delivered orders exceeded their promised delivery timestamp by strictly more than 24 hours. The tool reports facts; it does not decide compensation, contact customers or change orders. Proposed owner roles are not named people or confirmed appointments.

CTX2 - Trigger, scope and completion

Trigger, scope and completion: A manual CLI or existing local scheduler invokes one local batch run with two CSV paths and optional report date. Default is the previous UTC date. Join uses exact order_id equality; no normalization or fuzzy match. Success means one durable date directory with report.csv and a checksum-verified COMPLETE manifest. A valid empty day produces a header-only CSV. Non-goals: emails, messaging, remote updates, forecasting, chat UI, LLMs, customer actions and automatic correction of prior reports.

CTX3 - Date and data rules

Date and data rules: Validate entire snapshots before filtering. Blank actual timestamps are undelivered and excluded. A delivered actual ID with no promised row fails the run, even outside the report window; promised-only rows are counted as undelivered/not matched. Duplicate IDs in either file fail the entire run even if rows are identical. Only actual rows within [D 00:00:00Z,(D+1) 00:00:00Z) are reported. Signed lateness uses exact integer microseconds: flag iff actual-promised > 86400000000. Actual before promised is valid and not late. Empty values, bad headers/CSV/UTF-8, duplicate header names, invalid timestamps, non-UTC values and unsafe IDs block publication.

CTX4 - Output and local authority

Output and local authority: CSV columns in order: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late. Sort by case-sensitive ASCII order_id, preserve leading zeros, normalize timestamps to UTC Z with six fractional digits, emit signed decimal seconds with exactly six decimals and lowercase true/false. UTF-8 without BOM, CRLF records, standard CSV escaping. Local manifest includes schema_version, report_date, config_digest, input SHA-256 hashes, row counts, excluded-undelivered/out-of-window counts, report SHA-256, status COMPLETE and run_key. No raw rows in logs. This design exercise writes only design artifacts; running or scheduling the business report is separate implementation work. Manifest is exactly {schema_version:string="delivery-report-v1",report_date:YYYY-MM-DD,config_digest:hex64,promised_sha256:hex64,actual_sha256:hex64,raw_count_promised:int>=0,raw_count_actual:int>=0,rows:int>=0,flagged_rows:int>=0,excluded_undelivered:int>=0,excluded_out_of_window:int>=0,promised_only_rows:int>=0,report_sha256:hex64,status:string="COMPLETE",run_key:hex64}; no timestamp or absolute path is included in this deterministic object. Serialize manifest JSON with sorted keys and compact separators in UTF-8 followed by one LF. config_digest = SHA256 of compact sorted-key UTF-8 JSON with exactly {schema_version,threshold_microseconds,max_rows,max_bytes_per_file,deadline_seconds}; parser/ID/output contracts are versioned by schema_version; allowlisted paths are deployment capabilities, excluded from digest. run_key = SHA256 of compact sorted-key UTF-8 JSON with exactly {schema_version,report_date,config_digest,promised_sha256,actual_sha256}. No LF in either hashed JSON input. Fields use the typed values in C1/C2/C3, not stringified numbers.

CTX5 - Quality and value measurements

Quality and value measurements: R1 defines daily record accuracy, publication completeness and operations review-time KPIs. Baselines are unknown. Hypothesis: exact deterministic fixtures pass 100%; all published files have valid manifests; reviewer effort decreases without any missed/extra flagged rows. Actual business impact requires future shadow/manual comparison and the R5 experiment; no savings or production accuracy is claimed.

Evidence boundary: brief quotes and proposed H assumptions are separate. Validation checks source form; it does not establish policy truth or permissions.

2 Task map

T0 - Produce the daily operations CSV

Tree position: Overall job / sequence / risk low

Acquire and snapshot -> validate -> join/classify -> publish -> record a local outcome. On failure or cancellation stop publication and emit an owned local outcome.

Resolution basis: Acquire and snapshot -> validate -> join/classify -> publish -> record a local outcome. On failure or cancellation stop publication and emit an owned local outcome.

Autonomy: Deterministic local code; no LLM or external action. The operator supplies permitted local paths; business implementation is not executed by this design exercise.

Frequency / examples: Once per UTC day proposed in H5; bounded manual backfill allowed. / T1 then T2 then T3 then T4 then T5

T1 - Acquire the local run and immutable input bytes

Tree position: Child of T0 / leaf / risk medium

Resolve configured paths and permissions; derive UTC window, reject current/future date, acquire exclusive per-date directory lock, bound sizes and read immutable bytes. Record hashes after reading; do not reread changing files.

Resolution basis: Resolve configured paths and permissions; derive UTC window, reject current/future date, acquire exclusive per-date directory lock, bound sizes and read immutable bytes. Record...

Autonomy: Deterministic local code; no LLM or external action. The operator supplies permitted local paths; business implementation is not executed by this design exercise.

Frequency / examples: Once per UTC day proposed in H5; bounded manual backfill allowed. / 2026-10-27 selects [2026-10-27T00:00:00Z,2026-10-28T00:00:00Z).

T2 - Validate both CSV snapshots

Tree position: Child of T0 / leaf / risk high

Parse the entire files using H1/H4, enforce required unique headers and ID uniqueness and ensure delivered actual IDs have promised rows. Reject all malformed or contradictory inputs with exact reason and row number.

Resolution basis: Parse the entire files using H1/H4, enforce required unique headers and ID uniqueness and ensure delivered actual IDs have promised rows. Reject all malformed or contradictory...

Autonomy: Deterministic local code; no LLM or external action. The operator supplies permitted local paths; business implementation is not executed by this design exercise.

Frequency / examples: Once per UTC day proposed in H5; bounded manual backfill allowed. / Duplicate O100 in actual returns DUPLICATE_ID and no CSV.

T3 - Join delivered orders and compute strict lateness

Tree position: Child of T0 / leaf / risk high

Exact string join, then daily-window filtering; compute signed microseconds without floats; build sorted rows, counts and deterministic CSV bytes.

Resolution basis: Exact string join, then daily-window filtering; compute signed microseconds without floats; build sorted rows, counts and deterministic CSV bytes.

Autonomy: Deterministic local code; no LLM or external action. The operator supplies permitted local paths; business implementation is not executed by this design exercise.

Frequency / examples: Once per UTC day proposed in H5; bounded manual backfill allowed. / Actual 2026-10-27T12:00:00.000001Z minus promised 2026-10-26T12:00:00Z is late; exactly 24h is false.

T4 - Publish a complete immutable report

Tree position: Child of T0 / leaf / risk high

Under the date lock, compare any existing completed date directory. Stage CSV and COMPLETE manifest, fsync files and staging directory, then rename date directory and fsync parent. Reconcile uncertain publication before any retry.

Resolution basis: Under the date lock, compare any existing completed date directory. Stage CSV and COMPLETE manifest, fsync files and staging directory, then rename date directory and fsync...

Autonomy: Deterministic local code; no LLM or external action. The operator supplies permitted local paths; business implementation is not executed by this design exercise.

Frequency / examples: Once per UTC day proposed in H5; bounded manual backfill allowed. / A crash after directory rename is reconciled as COMPLETE if final hashes match.

T5 - Record completion, cancellation or owned failure

Tree position: Child of T0 / leaf / risk medium

Write local status safely; release acquired lock on normal exit. Operations inspects the local result each day. Failure gets proposed owner, acknowledgement deadline, next action and visible state. Unknown publication and stale locks require manual inspection.

Resolution basis: Write local status safely; release acquired lock on normal exit. Operations inspects the local result each day. Failure gets proposed owner, acknowledgement deadline, next...

Autonomy: Deterministic local code; no LLM or external action. The operator supplies permitted local paths; business implementation is not executed by this design exercise.

Frequency / examples: Once per UTC day proposed in H5; bounded manual backfill allowed. / BLOCKED/INPUT_INVALID: owner fixes CSV export and reruns; no final report is produced.

Attribute	Tasks	Authority / loading
A1 Immutable CSV evidence	T1, T2, T3	Two operator-provided local CSV files; exact content hashes produced by C1. / session start

Attribute	Tasks	Authority / loading
A2 Run policy and report identity	T1, T3, T4	Versioned local config: schema_version=delivery-report-v1, threshold_microseconds=86400000000, UTC report date, allowlisted roots and H1/H4 parser profile; no live remote policy. / session start
A3 Durable publication and failure state	T4, T5	Dedicated local output root, per-date lock, staging directory, committed manifest/CSV and status record; C4/C5 produce record evidence. / session start

3 Components

Contracts below retain field names and action boundaries. Long values are compacted; canonical design.json remains authoritative.

C1 - Run controller and immutable read

Group / tasks: Code: tools / T1

CLI run --promised PATH --actual PATH [--report-date D] --out ROOT. Reject future/current day; fail rather than use partial reads. Before and after reading verify same inode/size/mtime_ns; change gives INPUT_IO. Hash exact bytes in memory. Lock mkdir is exclusive; never break an existing lock automatically. SHA256 run_key is hash of compact sorted-key UTF-8 JSON {schema_version,report_date,config_digest,promised_sha256,actual_sha256}; canonical JSON has no spaces. config_digest hashes parser/limits/output/threshold settings, excluding absolute paths. Cancellation checked at bounded row batches and immediately before publication; lock metadata contains pid, start time and run key when known. Whole-run deadline 120s; no read retries. Exact digest key sets/serialization are CTX4; schema_version versions all parser/ID/output behavior....

Inputs

command: required object {promised_path:absolute path,actual_path:absolute path,output_root:absolute path,report_date:YYYY-MM-DD optional default yesterday UTC}

config: required immutable {schema_version:string,threshold_microseconds:int=86400000000,allowlisted_roots:array[path],max_rows:int=100000,max_bytes_per_file:int=52428800,deadline_seconds:int=120}

clock.now_utc:required aware UTC datetime; signal.cancel_requested:required boolean, initially false

Outputs

outcome:enum READY|BLOCKED|BUSY|CANCELLED; reason:enum NONE|INVALID_DATE|PATH_DENIED|INPUT_IO|FILE_TOO_LARGE|OUTPUT_IO|LOCK_BUSY|CANCELLED|TIMEOUT

snapshot when READY:required {promised_bytes:bytes,actual_bytes:bytes,promised_sha256:hex64,actual_sha256:hex64,config_digest:hex64,run_key:hex64,report_date:date,window_start:UTC datetime,window_end:UTC datetime,lock_path:path,deadline:monotonic time}

lock_acquired:required boolean; retry_count:required int=0; error:optional {file_role:enum promised|actual|output,row_number:optional int}; no row contents

Why code: Dates, hashes, identity and locks are exact deterministic operations; no natural-language interpretation is needed.

Authorization / approval boundary: Operator-configured local filesystem permission only; no runtime approval conversation. Recheck local path/capability before reads/writes. No external action is authorized by this design.

C2 - CSV integrity validator

Group / tasks: Code: validation / T2

CSV parser strict mode, utf-8-sig decode and exact header names; header order may vary and extra columns are ignored, but repeated headers/row-width mismatches fail. Reject all IDs outside H1 and duplicate IDs including identical rows. Validate all promised timestamps and nonblank actual timestamps before filtering. Blank actual is null, not parse failure. Delivered actual IDs absent in promised yield MISSING_PROMISE. Promised-only rows are allowed. Invalid time/date/precision => INVALID_TIMESTAMP; parsed nonzero offset => NON_UTC. No normalization of IDs. All errors terminate the batch; no fallback parser or row skipping. Zero retries.

Inputs

snapshot:required C1 READY snapshot

cancel_requested:required boolean; deadline:required monotonic time

Outputs

outcome:enum VALID|INVALID|CANCELLED; reason:enum NONE|CSV_PARSE|ENCODING|MISSING_HEADER|DUPLICATE_HEADER|ROW_SHAPE|ROW_LIMIT|INVALID_ID|DUPLICATE_ID|MISSING_PROMISE|INVALID_TIMESTAMP|NON_UTC|CANCELLED|TIMEOUT

validated when VALID:required {promised:map[string,int microseconds since epoch],actual:map[string,int microseconds since epoch|null],raw_count_promised:int,raw_count_actual:int}

error when INVALID:required {reason:enum as above,file_role:enum promised|actual,row_number:int}; row 1 means header; no raw field value

Why code: Typed parsing and one-to-one constraints protect join identity and exact threshold arithmetic.

Authorization / approval boundary: Operator-configured local filesystem permission only; no runtime approval conversation. Recheck local path/capability before reads/writes. No external action is authorized by this design.

C3 - Exact join and renderer

Group / tasks: Code: validation / T3

For each actual-map ID with nonnull timestamp use matching promised timestamp; include iff start<=actual<end. Calculate signed delta as int microseconds; flag delta>86400000000. Decimal seconds formatting has exactly six digits, including negative values; never round before comparison. Canonical timestamps use six fractional digits and Z. Sort ASCII order IDs, UTF-8 no BOM and CRLF including final record; all columns deterministic. Blank-actual records count as excluded_undelivered; promised-only_rows counts promised IDs absent from actual map. Empty valid input produces just the header. Check deadline/cancel every 1000 rows. No retries.

Inputs

validated:required C2 VALID maps

window_start/window_end:required integer UTC epoch microseconds from C1

threshold_microseconds:required int=86400000000

cancel_requested:required boolean; deadline:required monotonic time

Outputs

outcome:enum BUILT|CANCELLED; reason:enum NONE|CANCELLED|TIMEOUT

report when BUILT:required {csv_bytes:bytes,report_sha256:hex64,rows:int,flagged_rows:int,excluded_undelivered:int,excluded_out_of_window:int,promised_only_rows:int}

Why code: A code-only join and integer arithmetic are simpler and testable at the microsecond boundary.

Authorization / approval boundary: Operator-configured local filesystem permission only; no runtime approval conversation. Recheck local path/capability before reads/writes. No external action is authorized by this design.

C4 - Atomic local publisher

Group / tasks: Code: tools / T4

Final directory ROOT/YYYY-MM-DD contains report.csv and manifest.json. Under acquired per-date lock, existing COMPLETE manifest with same run_key and matching CSV hash => ALREADY_COMPLETE; any differing or incomplete final directory => CONFLICT. Stage at ROOT/.staging-<date>-<run_key>-<pid>-a<retry_count>; create files exclusively, fsync each and directory. Manifest schema is in CTX4; input counts are C2/C3 outputs. Recheck output-root permission and lock ownership before rename. Publish by same-filesystem directory rename then fsync output root; any exception after rename begins => UNKNOWN and C5 reconciliation, never immediate retry. Before rename, cancellation deletes staging and returns CANCELLED; failures known before rename return NOT_COMMITTED. After rename begins defer cancellation until reconciliation determines whether...

Inputs

snapshot:required C1 snapshot; report:required C3 BUILT report

retry_count:required int in 0..1; cancel_requested:required boolean

capability:required output-root permission from local execution account, rechecked before staging and rename; date lock still held

Outputs

outcome:enum COMMITTED|ALREADY_COMPLETE|NOT_COMMITTED|UNKNOWN|CONFLICT|CANCELLED; reason:enum NONE|PERMISSION_DENIED|STAGE_IO|PUBLISH_AMBIGUOUS|EXISTING_CONFLICT|CANCELLED|TIMEOUT

proof when COMMITTED or ALREADY_COMPLETE:required {final_dir:absolute path,manifest_path:absolute path,run_key:hex64,report_sha256:hex64,manifest_status:COMPLETE}

candidate_final_dir:required absolute path; cleanup_warning:optional enum STAGING_CLEANUP_FAILED|LOCK_CLEANUP_FAILED

Why code: Atomic publication with content identity prevents partial reports, duplicate runs and false completion.

Authorization / approval boundary: Operator-configured local filesystem permission only; no runtime approval conversation. Recheck local path/capability before reads/writes. No external action is authorized by this design.

C5 - Read-only publication reconciler

Group / tasks: Code: validation / T4

Read final manifest and CSV under held lock. COMPLETE + expected run_key + actual CSV SHA256 match => COMMITTED. No final directory => ABSENT; readable nonmatching/incomplete final => CONFLICT; read failure or reconciliation deadline => UNKNOWN. One reconciliation attempt, no write. ABSENT may authorize one C4 retry only if retry_count==0 and cancellation/deadline have not fired; increment before retry. Any unresolved ambiguity blocks owned outcome; an expired caller or stale lock never authorizes another publication. Matching proof also requires every expected manifest field from CTX4 (date/schema/config/input hashes and counts), not only status/run_key/output hash; mismatch yields CONFLICT. Read-only C5 never repairs or deletes a final directory.

Inputs

candidate_final_dir:required absolute path; expected_run_key:required hex64; expected_report_sha256:required hex64

read_capability:required within configured output root; retry_count:int 0..1; reconcile deadline:monotonic time now+5s

Outputs

outcome:enum COMMITTED|ABSENT|CONFLICT|UNKNOWN; reason:enum NONE|FINAL_ABSENT|FINAL_MISMATCH|READBACK_IO|TIMEOUT

proof when COMMITTED:required same proof shape as C4; observed_final_state:required enum MATCH|ABSENT|MISMATCH|UNREADABLE

Why code: Read-only evidence distinguishes a committed report from a safe retry and from unresolved ambiguity.

Authorization / approval boundary: Operator-configured local filesystem permission only; no runtime approval conversation. Recheck local path/capability before reads/writes. No external action is authorized by this design.

C6 - Local outcome recorder

Group / tasks: Code: human triggers / T5

Append a unique per-invocation local JSON status under ROOT/status (exclusive create + fsync), storing report date, run_key when known, terminal/reason, proof, proposed owner, ack_due_utc and manual next action. COMPLETE exits 0 only with C4/C5 proof; BLOCKED exits 2, CANCELLED exits 130. If status write fails emit structured stderr and status_proof=STDERR_ONLY; this does not convert a proved committed report to failure. Remove owned lock at normal exit; cleanup failure is visible stderr/status warning. Hard crash may leave stale lock: operator confirms no running process and read-only checks final manifest before manual lock removal. Operations checks local statuses by 10:00 UTC, acknowledges failure in a separate local operating log by 12:00 UTC, then manually escalates to lead if overdue. No automatic messaging and no indefinite...

Inputs

terminal:required enum COMPLETE|BLOCKED|CANCELLED

reason:required enum READY|ALREADY_COMPLETE|INPUT_INVALID|INPUT_IO|OUTPUT_IO|BUSY|CONFLICT|PUBLICATION_UNKNOWN|TIMEOUT|CANCELLED

proof:required on COMPLETE, otherwise absent; lock_acquired:required bool

owner_role:required proposed Operations report owner; acknowledgement_due:required next 12:00 UTC following this run

Outputs

state:enum COMPLETE|BLOCKED|CANCELLED; exit_code:int enum 0|2|130; status_path:optional absolute path

status_proof:enum DURABLE|STDERR_ONLY; lock_status:enum RELEASED|NOT_HELD|CLEANUP_FAILED

Why code: A file/exit-code handoff handles exceptions without adding a conversational or remote interface.

Authorization / approval boundary: Operator-configured local filesystem permission only; no runtime approval conversation. Recheck local path/capability before reads/writes. No external action is authorized by this design.

4 Workflow map

Node / component	To	Guard or terminal disposition
W1 / C1	W2	C1.outcome == READY [exclusive]
W1 / C1	W10	C1.outcome == CANCELLED [exclusive]
W1 / C1	W11	C1.outcome in [BLOCKED,BUSY] [exclusive]
W2 / C2	W3	C2.outcome == VALID [exclusive]
W2 / C2	W10	C2.outcome == CANCELLED and C2.reason == CANCELLED [exclusive]
W2 / C2	W11	C2.outcome == INVALID or (C2.outcome == CANCELLED and C2.reason == TIMEOUT) [exclusive]
W3 / C3	W4	C3.outcome == BUILT [exclusive]
W3 / C3	W10	C3.outcome == CANCELLED and C3.reason == CANCELLED [exclusive]
W3 / C3	W11	C3.outcome == CANCELLED and C3.reason == TIMEOUT [exclusive]
W4 / C4	W9	C4.outcome in [COMMITTED,ALREADY_COMPLETE] [exclusive]
W4 / C4	W6	C4.outcome == UNKNOWN [exclusive]
W4 / C4	W10	C4.outcome == CANCELLED [exclusive]
W4 / C4	W11	C4.outcome in [NOT_COMMITTED,CONFLICT] [exclusive]
W6 / C5	W9	C5.outcome == COMMITTED [exclusive]
W6 / C5	W4	C5.outcome == ABSENT and retry_count == 0 and not cancel_requested and not deadline_expired; increment retry_count before invoking C4 [exclusive]
W6 / C5	W10	C5.outcome == ABSENT and cancel_requested [exclusive]
W6 / C5	W11	C5.outcome in [CONFLICT,UNKNOWN] or (C5.outcome == ABSENT and not cancel_requested and (retry_count == 1 or deadline_expired)) [exclusive]
W9 / C6	TERMINAL	COMPLETE with required proof; C6 status write failure yields STDERR_ONLY and a warning; never invents a report proof.
W10 / C6	TERMINAL	CANCELLED; final absent has been proved if publication was attempted; owned lock released where possible.
W11 / C6	TERMINAL	BLOCKED with reason and proposed owner; exit 2. Publication ambiguity retains run identity for manual read-only inspection and forbids another blind write.

Failure,retry and human controls

failureDisposition [W1, W2, W3, W4, W6, W9, W10, W11]: All outcomes are enumerated below; unhandled internal exceptions go to W11 as BLOCKED/OUTPUT_IO without a success claim.

timeout [W1, W2, W3, W4, W6, W9, W10, W11]: C1-C4 share C1.deadline (120s); C5 has one additional 5s read-only reconciliation budget; C6 attempts local status once then stderr.

retryPolicy [W1, W2, W3, W4, W6, W9, W10, W11]: Zero except W6 -> W4 after ABSENT, maximum one retry, bounded by deadline and cancellation.

5 Framework, tools and data

F1 - Code workflow fit

Category / permission: fit / none

Chosen design: A deterministic Python CLI is the complete first release; zero agent framework and zero model calls.

Rejected alternatives: Rejected LLM/multi-agent system: adds parsing uncertainty, credentials, cost and orchestration with no judgement task.

Tradeoffs / verification: Both options assessed on exact join/date correctness, local deployment, retries/audit and operating burden; code directly satisfies every criterion. No vendor capability claim is needed.

Blast radius: No external systems.

Authority: Use locally granted path capabilities; no account connection or remote write. Proposed owners confirm launch prerequisites, not this design review.

Least privilege: Read-only configured input roots; create/write/delete staging and own lock/status only inside dedicated output root; never alter inputs or existing committed date reports. No network capability or secrets.

aws: If future hosting is requested: private object/file storage plus a scheduled container, unverified candidate requiring equivalent immutable-pair, atomic-publication, permission and retention tests; no current integration.

gcp: If future hosting is requested: private object/file storage plus a scheduled container, unverified candidate on the same criteria; no current integration.

azure: If future hosting is requested: private object/file storage plus a scheduled container, unverified candidate on the same criteria; no current integration.

openSource: Chosen: Python 3 standard library csv, datetime, decimal/integer formatting, hashlib, json, pathlib/os; local filesystem. Existing local scheduler optional after deployment verification; no paid SDK or LLM service.

F2 - One-host durable batch architecture

Category / permission: architecture / write

Chosen design: C1 -> C2 -> C3 -> C4 with C5 reconciliation and C6 local terminal status. Per-date exclusive lock, content-bound run identity and immutable report directory.

Rejected alternatives: Rejected database/workflow service for this small local report: extra infrastructure and operational state without a remote or long-running approval requirement.

Tradeoffs / verification: Local filesystem behavior is a launch dependency; a future distributed execution would require a new concurrency/publication design.

Blast radius: Only the configured local report root.

Authority: Use locally granted path capabilities; no account connection or remote write. Proposed owners confirm launch prerequisites, not this design review.

Least privilege: Read-only configured input roots; create/write/delete staging and own lock/status only inside dedicated output root; never alter inputs or existing committed date reports. No network capability or secrets.

Cloud candidate details match F1.

F3 - Schedule and bounded orchestration

Category / permission: orchestration / none

Chosen design: Invoke CLI manually first; use an existing local daily scheduler at proposed 08:00 UTC after export availability is verified. One run bounded to 120s plus 5s ambiguous-commit readback.

Rejected alternatives: Rejected cron-like remote service provisioning: outside design scope and not necessary for a small local job.

Tradeoffs / verification: Manual invocation and an existing scheduler have the same validation, locks, exact data and publication rules; scheduling only changes trigger ownership. No automation is created in this

exercise.

Blast radius: One run for one completed UTC date.

Authority: Use locally granted path capabilities; no account connection or remote write. Proposed owners confirm launch prerequisites, not this design review.

Least privilege: Read-only configured input roots; create/write/delete staging and own lock/status only inside dedicated output root; never alter inputs or existing committed date reports. No network capability or secrets.

Cloud candidate details match F1.

F4 - Filesystem authority and inert inputs

Category / permission: permission / write

Chosen design: Treat CSV content as untrusted data; strict parser/ID profile, no eval or shell interpolation. Resolve paths against allowlisted roots, forbid symlinks, fresh capability checks and exclusive creation.

Rejected alternatives: Rejected processing arbitrary paths and preserving formula-like IDs directly in spreadsheet CSV: unclear scope and formula risk.

Tradeoffs / verification: The narrow ID profile is a confirmable source-contract dependency. Wrong-namespace/stale snapshot risk still needs the source-owner procedure; hashes establish byte identity, not business provenance.

Blast radius: Input bytes read only; local output and status contain order identifiers and timestamps.

Authority: Use locally granted path capabilities; no account connection or remote write. Proposed owners confirm launch prerequisites, not this design review.

Least privilege: Read-only configured input roots; create/write/delete staging and own lock/status only inside dedicated output root; never alter inputs or existing committed date reports. No network capability or secrets.

Cloud candidate details match F1.

F5 - Minimized data and retention

Category / permission: data / write

Chosen design: Memory-only snapshots; retain output CSV, hashes, count/manifest and error codes/row numbers locally for proposed 30 days. No customer names, addresses or raw rows in logs.

Rejected alternatives: Rejected cloud warehouse/vector store: no retrieval or analytics need and extra data movement.

Tradeoffs / verification: Memory/row limits are provisional; real export profiling and retention approval remain launch gates. All cloud alternatives face the same access, schema, provenance, durability, deletion and observability criteria; no price or reliability assumptions are asserted.

Blast radius: A single operator-controlled filesystem; no upload.

Authority: Use locally granted path capabilities; no account connection or remote write. Proposed owners confirm launch prerequisites, not this design review.

Least privilege: Read-only configured input roots; create/write/delete staging and own lock/status only inside dedicated output root; never alter inputs or existing committed date reports. No network capability or secrets.

Cloud candidate details match F1.

6 Models and cost

No LLM components are included. No model plan or provider cost is required.

7 Failure modes

Risk	Specific failure	S / O / D	RPN	Linked evaluations
FM1	Wrong or duplicate order identity	7 / 3 / 3	63	EV1, EV3, EV4, EV5, EV30, EV31, EV32
FM2	Threshold or UTC-window error	8 / 3 / 4	96	EV1, EV2, EV6, EV30, EV31, EV32
FM3	Incomplete, malformed or stale evidence	8 / 4 / 4	128	EV7, EV8, EV9, EV10, EV11, EV12, EV19, EV22, EV30, EV31, EV32
FM4	Partial or false-complete local publication	8 / 3 / 4	96	EV13, EV14, EV15, EV16, EV20, EV23, EV30, EV31, EV32
FM5	Concurrent or conflicting re-publication	7 / 3 / 3	63	EV17, EV18, EV24, EV25, EV30, EV31, EV32
FM6	Unsafe input or filesystem scope	8 / 2 / 4	64	EV5, EV20, EV21, EV30, EV31, EV32
FM7	Cancellation, deadline or size-limit mishandling	7 / 3 / 3	63	EV16, EV22, EV26, EV27, EV30, EV31, EV32
FM8	Failure is not owned or seen	6 / 4 / 4	96	EV28, EV30, EV31, EV32
FM9	Exposure or uncontrolled retention	6 / 3 / 4	72	EV29, EV30, EV31, EV32

Scores and RPN are author-supplied prioritization estimates,not incident probabilities or verified risk acceptance.

FM1 - Wrong or duplicate order identity

Effect / cause: Rows joined to wrong orders or duplicate report rows. / Padded/normalized IDs, cross-namespace exports or duplicate IDs.

Prevent / detect: Exact case-sensitive H1 ID validation, fail duplicate IDs; same-namespace source procedure H2. / Validator errors plus manual source-pair checklist before launch.

Owned action / gate: Data owner must approve H1/H2; release EV1/EV3/EV4/EV5 and source-pair runbook evidence before launch.

Metric / business KPI: wrong-join/duplicate rows per published report; target 0 / R1 record accuracy

FM2 - Threshold or UTC-window error

Effect / cause: Late order omitted or exactly-24h order incorrectly flagged; wrong daily date included. / Float rounding, >= instead of >, local timezone conversion or inclusive end boundary.

Prevent / detect: Parse UTC exact microseconds, strict >, half-open UTC day and sorted deterministic output. / Hand-worked microsecond boundary goldens and independent manual oracle.

Owned action / gate: Implementation owner blocks release until EV1/EV2/EV6 pass exactly; no averaging across boundary failures.

Metric / business KPI: incorrect flags or window inclusions / expected delivered rows / R1 record accuracy

FM3 - Incomplete, malformed or stale evidence

Effect / cause: A plausible but incomplete report hides late deliveries. / Broken CSV/encoding, missing promise, malformed/non-UTC time, file changes during read or stale exports.

Prevent / detect: Whole-file validation and immutable snapshot hash; fail missing delivered promise; H2 source cutoff procedure. No inferred promise or silent skip. / Reject exact parse/relationship reason; compare source counts and export cutoff manually.

Owned action / gate: Data owner must verify source freshness/late-arrival procedure H2 and implementation owner must pass EV7-EV12, EV19, EV22 before launch.

Metric / business KPI: published reports with invalid/unverified inputs / runs; target 0 / R1 record accuracy and R2 completeness

FM4 - Partial or false-complete local publication

Effect / cause: Operations consumes a partial file or job claims completion without durable proof. / Disk-full, revoked write access, crash/timeout around rename, readback failure.

Prevent / detect: Stage and fsync before atomic directory publication; manifest+hash required; read-only reconciliation before one bounded retry; unknown stays blocked. / Fault-injected staging, rename and readback; checksum audit for every completed report.

Owned action / gate: Implementation owner owns filesystem verification H6 and release EV13-EV16/EV20/EV23 before launch; unresolved ambiguity blocks consumption.

Metric / business KPI: complete statuses lacking matching final proof / complete statuses; target 0 / R2 publication completeness

FM5 - Concurrent or conflicting re-publication

Effect / cause: Multiple versions for one date confuse operations or overwrite valid output. / Overlapping schedule, rerun with changed bytes or stale lock after crash.

Prevent / detect: Per-date exclusive lock; same input+config no-op; changed existing date conflicts; never auto-break lock or auto-replace. / Concurrent command and same/different-content rerun tests; local status shows BUSY/CONFLICT.

Owned action / gate: Report owner must approve correction/stale-lock runbook H8 and pass EV17/EV18/EV24/EV25 before launch.

Metric / business KPI: duplicate/overwritten date reports; target 0 / R2 publication completeness

FM6 - Unsafe input or filesystem scope

Effect / cause: Order cells execute as spreadsheet formulas or files escape allowed roots. / Formula-prefixed IDs, shell evaluation, symlink paths, permissions revoked after validation.

Prevent / detect: H1 safe ID profile; parse as inert CSV with no shell/eval; allowlisted real paths and no symlinks; fresh local permission check before publication. / Injection/path/revoked-permission isolated goldens and capability audit.

Owned action / gate: Implementation owner must pass EV5/EV20/EV21 and approve deployment H6; strict ID profile may be revised only with equivalent safe CSV encoding tests.

Metric / business KPI: unsafe/unauthorized output files; target 0 / R1 record accuracy plus privacy guardrail

FM7 - Cancellation, deadline or size-limit mishandling

Effect / cause: Unbounded runtime or a cancelled job leaves a consumable incomplete output. / Large export, ignored signal, cancellation during rename or blind retry after timeout.

Prevent / detect: Byte/row/time bounds, cancellation before rename and deferred outcome reconciliation after rename begins; no blind retry. / Synthetic deadline/size/signal fault tests assert terminal and final proof.

Owned action / gate: Implementation owner passes EV16/EV22/EV26/EV27 before launch and measures actual budget H5.

Metric / business KPI: runs beyond configured bound or unproved cancellation statuses / R2 publication completeness and R3 review effort

FM8 - Failure is not owned or seen

Effect / cause: Report is missing while operations assumes the day was processed. / Scheduler failure, status file unavailable, stale lock or no assigned human acknowledgement.

Prevent / detect: Exit code + local durable status, stderr fallback, daily inspection and manual acknowledgement/escalation in H5; never send email or remote updates. / Missing-date/status count check and local operating log review; status-I/O fault test.

Owned action / gate: Operations report owner confirms staffing/deadlines H5 and passes EV28; operations lead reviews any missed acknowledgement.

Metric / business KPI: unacknowledged blocked dates past 12:00 UTC / blocked dates / R2 completeness and R3 review effort

FM9 - Exposure or uncontrolled retention

Effect / cause: Unneeded order data is exposed or retained beyond agreed policy. / Raw rows in error logs, overly broad directory access or no approved cleanup practice.

Prevent / detect: Persist minimized columns/hashes/counts; error row numbers only; dedicated private local root; H7 approval and manual retention procedure. / Inspect output/status contents and filesystem access/retention checklist.

Owned action / gate: Data owner approves retention/access H7 and release EV29 before launch; implementation does not auto-delete reports.

Metric / business KPI: raw-input payloads in status/logs and expired artifacts; target 0 after approved retention / Privacy guardrail supporting R2 trustworthy report use

8 Evaluations

These are test specifications,not executed results. Full self-contained fixtures,ordered mocks and inheritance rules are in eval-set.json and design.json. This report shows concrete outcomes and assertion summaries.

EV1 - Positive one-row late delivery

Layer / links: golden / tasks T1, T2, T3, T4, T5 / risks FM1, FM2

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; exit_code: 0; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late O1,2026-10-26T12:00:00.000000Z,2026-10-27T12:00:00.000001Z,86400.000001,true ; rows: 1; flagged_rows: 1

Assertion summary: all: csv bytes equal expected.csv_utf8 | C4 final report and COMPLETE manifest hashes match | one final publish and no input mutation

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV2 - Exactly 24 hours is not late

Layer / links: golden / tasks T3 / risks FM2

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late O1,2026-10-26T12:00:00.000000Z,2026-10-27T12:00:00.000000Z,86400.000000,false ; flagged_rows: 0

Assertion summary: all: csv bytes equal expected.csv_utf8 | boolean is false, not >= threshold

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV3 - Duplicate promised ID fails even with identical rows

Layer / links: golden / tasks T2 / risks FM1

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1;

threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: DUPLICATE_ID; file_role: promised; row_number: 3

Assertion summary: all: C2.reason == DUPLICATE_ID | error file and row equal expected | no final report or C3/C4 invocation

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV4 - Duplicate actual ID fails

Layer / links: golden / tasks T2 / risks FM1

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: DUPLICATE_ID; file_role: actual; row_number: 3

Assertion summary: all: C2.reason == DUPLICATE_ID | error row_number == 3 | no final report or C3/C4 invocation

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV5 - Formula-like ID is inert and rejected

Layer / links: golden / tasks T1, T2, T4 / risks FM1, FM6

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: INVALID_ID; file_role: promised; row_number: 2

Assertion summary: all: C2.reason == INVALID_ID | no evaluation, shell invocation or network call | no final report

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV6 - UTC day, sorting, leading zero and early delivery

Layer / links: golden / tasks T3 / risks FM2

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; csv_utf8:
order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late
001,2026-10-26T00:00:00.000000Z,2026-10-27T00:00:00.000000Z,86400.000000,false
O2,2026-10-28T00:00:00.000000Z,2026-10-27T23:59:59.999999Z,-0.000001,false
; rows: 2; excluded_out_of_window: 1

Assertion summary: all: csv bytes equal expected.csv_utf8 | day start included; next midnight excluded | negative delta exact; 001 retains leading zeros

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV7 - Delivered order missing promise

Layer / links: golden / tasks T1, T2 / risks FM3

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: MISSING_PROMISE; file_role: actual; row_number: 2

Assertion summary: all: C2.reason == MISSING_PROMISE | exact error file/row match | no final report or row skipping

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV8 - Invalid actual date

Layer / links: golden / tasks T1, T2 / risks FM3

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: INVALID_TIMESTAMP; file_role: actual; row_number: 2

Assertion summary: all: C2.reason == INVALID_TIMESTAMP | exact error file/row match | no final report or row skipping

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV9 - Non-UTC actual offset

Layer / links: golden / tasks T1, T2 / risks FM3

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: NON_UTC; file_role: actual; row_number: 2

Assertion summary: all: C2.reason == NON_UTC | exact error file/row match | no final report or row skipping

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV10 - Missing required header

Layer / links: golden / tasks T1, T2 / risks FM3

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: MISSING_HEADER; file_role: promised; row_number: 1

Assertion summary: all: C2.reason == MISSING_HEADER | exact error file/row match | no final report or row skipping

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV11 - Duplicate header

Layer / links: golden / tasks T1, T2 / risks FM3

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: DUPLICATE_HEADER; file_role: promised; row_number: 1

Assertion summary: all: C2.reason == DUPLICATE_HEADER | exact error file/row match | no final report or row skipping

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV12 - CSV row width mismatch

Layer / links: golden / tasks T1, T2 / risks FM3

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: ROW_SHAPE; file_role: promised; row_number: 2

Assertion summary: all: C2.reason == ROW_SHAPE | exact error file/row match | no final report or row skipping

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV13 - Known pre-publication disk-full

Layer / links: golden / tasks T4 / risks FM4

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: OUTPUT_IO; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: STAGE_IO

Assertion summary: all: C4.outcome == NOT_COMMITTED | no final report and zero rename attempts | staging cleanup attempted; exit 2

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV14 - Unknown commit reconciles without duplicate publish

Layer / links: golden / tasks T4 / risks FM4

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; exit_code: 0; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late O1,2026-10-26T12:00:00.000000Z,2026-10-27T12:00:00.000001Z,86400.000001,true ; rename_attempts: 1; reconciliations: 1

Assertion summary: all: C4 UNKNOWN routes C5 once | C5 proof hashes derived from fixture match | no second rename; final CSV equals expected

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV15 - Proven absent permits one retry

Layer / links: golden / tasks T4 / risks FM4

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; rename_attempts: 2; retry_count: 1; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late O1,2026-10-26T12:00:00.000000Z,2026-10-27T12:00:00.000001Z,86400.000001,true

Assertion summary: all: C5 ABSENT precedes second C4 | retry_count incremented before retry | exactly one final directory and matching COMPLETE proof

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV16 - Unresolved publication is blocked

Layer / links: golden / tasks T1, T3, T4 / risks FM4, FM7

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: PUBLICATION_UNKNOWN; exit_code: 2; rename_attempts: 1; publication_state: UNKNOWN

Assertion summary: all: no retry after C5 UNKNOWN | no COMPLETE/CANCELLED claim without proof | local owned status retains run_key for manual inspection

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV17 - Concurrent run cannot write

Layer / links: golden / tasks T1, T4 / risks FM5

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: BUSY; exit_code: 2; final_dir_exists: false; write_count_final: 0

Assertion summary: all: C1.outcome == BUSY | no read/join/staging/publish after lock refusal | other process lock not removed

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV18 - Changed input never overwrites report

Layer / links: golden / tasks T1, T4 / risks FM5

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: CONFLICT; exit_code: 2; existing_report_unchanged: true; new_final_writes: 0

Assertion summary: all: C4.outcome == CONFLICT | existing bytes preserved | no rename or correction command

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV19 - Undelivered and promised-only produce empty valid CSV

Layer / links: golden / tasks T1, T2 / risks FM3

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late ; rows: 0; flagged_rows: 0; excluded_undelivered: 1; promised_only_rows: 1

Assertion summary: all: header-only CSV equals expected | blank actual excluded without invented timestamp | COMPLETE manifest counts match

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV20 - Revoked local write permission fails safely

Layer / links: golden / tasks T1, T2, T4 / risks FM4, FM6

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: OUTPUT_IO; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: PERMISSION_DENIED

Assertion summary: all: no staging or final write after denial | exit 2, stderr fallback permitted if status path also denied | no privilege escalation

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV21 - Symlink path cannot escape input scope

Layer / links: golden / tasks T1, T2, T4 / risks FM6

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: OUTPUT_IO; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: PATH_DENIED

Assertion summary: all: C1 blocks before input read | outside file is neither read nor written | no final report

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV22 - Input size exceeds configured bound

Layer / links: golden / tasks T1, T2, T3, T4 / risks FM3, FM7

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_IO; exit_code: 2; final_dir_exists: false; write_count_final: 0; component_reason: FILE_TOO_LARGE

Assertion summary: all: C1 rejects before reading oversized bytes | no truncation or final report

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV23 - Retry budget cannot loop

Layer / links: golden / tasks T4 / risks FM4

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: OUTPUT_IO; exit_code: 2; final_dir_exists: false; write_count_final: 0; rename_attempts: 2; retry_count: 1

Assertion summary: all: exactly two C4 attempts and two C5 readbacks | no third rename | final absent; BLOCKED/OUTPUT_IO

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV24 - Exact rerun returns durable no-op

Layer / links: golden / tasks T1, T4 / risks FM5

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; reason: ALREADY_COMPLETE; exit_code: 0; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late O1,2026-10-26T12:00:00.000000Z,2026-10-27T12:00:00.000001Z,86400.000001,true ; new_final_writes: 0

Assertion summary: all: verify stored report hash and run_key before ALREADY_COMPLETE | zero stage/rename calls | input/config hash equality, not filename equality

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV25 - Stale lock is never auto-broken

Layer / links: golden / tasks T1, T4 / risks FM5

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: BUSY; exit_code: 2; final_dir_exists: false; write_count_final: 0

Assertion summary: all: no automatic lock deletion or publication | local outcome names manual inspection owner | no unsupported claim that old process is dead

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV26 - Cancellation before publication

Layer / links: golden / tasks T1, T3, T4 / risks FM7

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: CANCELLED; reason: CANCELLED; exit_code: 130; final_dir_exists: false; write_count_final: 0

Assertion summary: all: staging cleanup attempted | zero rename calls; owned lock released when possible | local cancellation outcome; no success claim

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV27 - Cancellation after commit start defers to proof

Layer / links: golden / tasks T1, T3, T4 / risks FM7

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; exit_code: 0; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late O1,2026-10-26T12:00:00.000000Z,2026-10-27T12:00:00.000001Z,86400.000001,true ; rename_attempts: 1

Assertion summary: all: C4 returns UNKNOWN then C5 COMMITTED | published report proof takes precedence over cancellation | no compensation, deletion or second publication

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV28 - Local status failure remains visible

Layer / links: golden / tasks T5 / risks FM8

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: BLOCKED; reason: INPUT_INVALID; exit_code: 2; final_dir_exists: false; write_count_final: 0; status_proof: STDERR_ONLY; component_reason: DUPLICATE_ID; stderr_contains: BLOCKED | INPUT_INVALID | Operations report owner

Assertion summary: all: exit 2 with structured stderr owner and manual next action | no external messages | no completed report or false durable-status claim

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV29 - Extra data is minimized from outputs and logs

Layer / links: golden / tasks T4, T5 / risks FM9

Fixture: command: promised_path: /sandbox/in/promised.csv; actual_path: /sandbox/in/actual.csv; output_root: /sandbox/out; report_date: 2026-10-27; config: schema_version: delivery-report-v1; threshold_microseconds: 86400000000; allowlisted_roots: /sandbox/in; max_rows: 100000; max_bytes_per_file: 52428800; deadline_seconds: 120;...

Expected summary: terminal: COMPLETE; csv_utf8: order_id,promised_delivery_utc,actual_delivery_utc,lateness_seconds,is_more_than_24h_late O1,2026-10-26T12:00:00.000000Z,2026-10-27T12:00:00.000001Z,86400.000001,true ; forbidden_output_substrings: PRIVATE_MARKER_92 | customer_secret

Assertion summary: all: CSV equals expected exact five-column bytes | manifest/status/stderr contain no forbidden substring or raw row | no input copy, remote action or automatic retention deletion

Metric / target: Exact fixture assertion pass fraction / 100% of assertions; any failure blocks release

Labels / cadence / alert: Expected threshold and UTC boundary outputs hand-computed; second operations reviewer checks proposed fixture labels before launch. / Every relevant code/config change and before release; not run in this design exercise. / Any unexpected final CSV, wrong terminal/reason, incorrect flag or forbidden effect fails.

EV30 - Offline parser, invariants and end-to-end suite

Layer / links: offline / tasks T1, T2, T3, T4, T5 / risks FM1, FM2, FM3, FM4, FM5, FM6, FM7, FM8, FM9

Fixture: Run EV1-EV29 as isolated temporary-root end-to-end fixtures plus component/property cases for C1-C6. Add generated timestamp pairs around -1us/0/24h/+1us and midnight, empty files versus header-only files, 100000/100001 row bounds, malformed...

Expected summary: All exact goldens and component invariants pass. For every output row, recomputation from unique source rows agrees; counted partitions sum to actual-map rows; canonical output/hash stable under input row permutations. Duplicate and malformed data never publishes.

Assertion summary: Release failure if any golden fails; run at least all listed deterministic component classes, seeded property corpus with seed recorded and an independent reference join using Decimal/timedelta microseconds. Record actual counts/results, not just suite existence.

Metric / target: Exact assertion pass fraction; published-invalid-report count / 100%; zero invalid publication

Labels / cadence / alert: Implementer writes harness; operations reviewer verifies hand labels and real-source strata after H1/H2 confirmation. / Each code/config change and before any rollout phase / Any failed assertion blocks release; owner Implementation owner.

EV31 - Daily local report and failure monitor

Layer / links: online / tasks T1, T2, T3, T4, T5 / risks FM1, FM2, FM3, FM4, FM5, FM6, FM7, FM8, FM9

Fixture: For each expected UTC date inspect only local date directory/manifest/hash/status and operating log; compare approved export cutoff evidence and input counts. Review ID/flag anomalies against source. Check no raw data in logs, retention age and...

Expected summary: One valid completed report or explicit owned blocked/cancelled status per invocation; missing scheduled invocation visible by 10:00 UTC. Unknown/conflicting publication is never marked complete. Failures acknowledged by 12:00 UTC.

Assertion summary: Record count of missing dates, hash mismatches, unacknowledged blocks and invalid/extra/missing flags; any integrity/privacy violation stops consumption and owner investigates. Volume shifts are reviewed, not proof of incorrectness. Manual escalation only; no remote updates.

Metric / target: Complete dates / expected dates; checksum validity; sampled exact match; overdue ack count / Completeness target 100% (hypothesis), integrity breaches 0; baseline collection before target claims.

Labels / cadence / alert: Operations report owner checks manifest and source cutoff; implementation owner investigates integrity faults; no automated truth claim from hashes. / Every expected day by 10:00 UTC; ack check 12:00 UTC; weekly retention review. / Any wrong/missing/extra row or checksum mismatch stops use; missing date/overdue ack requires manual action.

EV32 - Release gate and deployment filesystem proof

Layer / links: release / tasks T1, T2, T3, T4, T5 / risks FM1, FM2, FM3, FM4, FM5, FM6, FM7, FM8, FM9

Fixture: Collect executed EV1-EV29 and EV30 result files, real-source H1/H2/H4 approval, verified H5 staffing/schedule/budget, H6 allowlist/lock/fsync/rename fault proof, H7 access/retention approval and H8 correction runbook. Run EV31 shadow monitor...

Expected summary: Versioned build/config hashes; all named suites pass and all launch dependencies have owner evidence; no unresolved integrity issue. Design review alone never satisfies this release gate.

Assertion summary: required_golden_ids: EV1 | EV2 | EV3 | EV4 | EV5 | EV6 | EV7 | EV8 | EV9 | EV10 | EV11 | EV12 | EV13 | EV14 | EV15 | EV16 | EV17 | EV18 | EV19 | EV20 | EV21 | EV22 | EV23 | EV24 | EV25 | EV26 | EV27 | EV28 | EV29; require_all_pass: true; required_additional_ids: EV30; required_assumption_evidence: H1 | H2 | H3 | H4 | H5 | H6 | H7 | H8; no_unresolved_integrity_findings: true; owner: Operations report owner and Implementation owner; live_checks: EV31 procedure demonstrated in shadow; ongoing results are not...

Metric / target: Release-gate checklist evidence completeness / 100% evidence; zero unresolved integrity failures

Labels / cadence / alert: Owners inspect actual result files and source-contract evidence; scores in review.json do not substitute. / Before shadow -> assist and every material code/config/parser change. / Any missing result, failed fixture or unverified launch blocker prevents operational consumption.

9 Measurement and rollout

R1 - Exact report accuracy and business evidence

Category / phase / owner: kpi / All phases / Operations data owner

Business outcome: Daily record accuracy = correctly included/flagged rows divided by independently expected rows; track missing and extra rows separately so denominator cannot hide omissions. Weekly aggregate is secondary to any integrity breach. Baseline unknown: manual source comparison in shadow. Target hypothesis: exact agreement 100%; any error blocks consumption.

AI mechanisms / guardrails: No AI metrics. Exact join/UTC fixture pass fraction causally protects correct queue selection by eliminating date/identity errors. Source freshness is separately required: correct arithmetic cannot repair omitted deliveries.

Entry / exit / stop and rollback: Do not declare production accuracy from synthetic EVs; record verified manual sample denominator and mismatch counts daily; all known integrity defects must be zero.

Experiment: R5 measures operations effort only after accuracy gates pass.

R2 - Completeness and durable report KPI

Category / phase / owner: kpi / All phases / Operations report owner

Business outcome: Publication completeness = expected report dates with one checksum-verified COMPLETE directory by 10:00 UTC / expected scheduled report dates in each UTC week; blocked/cancelled/missing dates remain in denominator. Baseline unknown; target hypothesis 100%.

AI mechanisms / guardrails: Guardrail: false-complete count must be 0; correct durable proof makes a report safely consumable. Busy/conflict/unknown counts diagnose reliability, not delivery performance.

Entry / exit / stop and rollback: Any invalid hash/false completion stops report use immediately, irrespective of average timeliness.

Experiment: Observe during shadow before proposing service level.

R3 - Manual review effort

Category / phase / owner: kpi / Shadow onward / Operations report owner

Business outcome: Operations review effort = minutes from export pair availability to verified daily queue, including correction/investigation time, per UTC report day. Baseline unknown; collect manual timers during shadow. Target hypothesis: reduce median time without any accuracy/completeness breach.

AI mechanisms / guardrails: Runtime p95 under provisional 120s and output verification time are operational metrics. Runtime affects availability but is not a substitute for correctness. Human ack rate is a guardrail for failures.

Entry / exit / stop and rollback: Do not claim savings before R5 measurement; safety/error gates are applied per day.

Experiment: R5 uses date-level paired observations and continuous-outcome power analysis.

R4 - Shadow then assist-only rollout

Category / phase / owner: phase / Shadow -> Assist-only -> Routine assist; no autonomous operations action phase. / Operations report owner owns stop/restore; Implementation owner repairs; Data owner confirms source truth.

Business outcome: R1/R2/R3 definitions above.

AI mechanisms / guardrails: EV1-EV30 executed pass; local monitor EV31 and release evidence EV32.

Entry / exit / stop and rollback: Shadow entry: implemented CLI/harness and EV32 local synthetic/dependency evidence complete; no user consumption. Shadow exit: five consecutive complete daily export pairs spanning approved source strata, exact manually adjudicated report agreement and no unknown publication, with H2 source cutoff checks. Assist entry: shadow exit signed locally by proposed owners. Assist exit: five additional days with checks and failure drill; R5 effort evidence may extend duration. Routine assist entry: owners approve runbook and monitoring. Any wrong row, privacy leak, invalid proof or unowned failure stops consumption; preserve artifacts, disable schedule manually and use current manual join procedure. Re-enable only after cause fixed and EV32 rerun.

Experiment: No expansion of side-effect authority at any phase. Shadow/assist day counts are proposed minimum coverage, not statistical power or observed data.

R5 - Measure effort with paired days

Category / phase / owner: experiment / During shadow/assist after deterministic release gates. / Operations report owner approves effect threshold; analyst/implementation owner computes sample and review.

Business outcome: R3 minutes per report day; R1/R2 integrity are non-negotiable guardrails.

AI mechanisms / guardrails: Paired difference in independently timed manual versus tool-assisted review minutes; medians and mean paired difference with confidence interval; no AI accuracy claim.

Entry / exit / stop and rollback: Eligibility: complete approved source pairs in the requested UTC day; strata by row-count bands and late-order prevalence measured from sources. Each day is one paired block.

Two separate operators review same snapshot independently; randomize method-to-operator order within day and rotate roles. Hide each other's result until timing/labels recorded; reconcile afterwards. Start with baseline collection across weekdays/volume strata, estimate paired-difference SD, choose minimum meaningful minutes reduction with owner, use paired continuous-mean power calculation for 80% power and two-sided alpha 0.05. Adjust for repeated operator/week clusters using estimated intracluster correlation or cluster bootstrap; report uncertainty when sample size is insufficient. Continue complete weekly cycles until precomputed sample and strata achieved; five-day phase minima are not a powered sample. Any integrity breach stops experiment use, not averaged away.

Experiment: Unit is UTC report date; no randomization of customer service. Operators must not share report outputs before timing. If independent operators unavailable, record limitation and run descriptive timings without causal savings claim.

R6 - Accountability and manual recovery

Category / phase / owner: owner / All phases / Proposed roles only; appointment and capacity require H5/H7/H8 evidence.

Business outcome: R1 source correctness: Data owner. R2 schedule/outcome/ack: Report owner. R3 timing: Report owner. Code/filesystem/control repairs: Implementation owner.

AI mechanisms / guardrails: Local status proof, overdue acknowledgements and source export checks.

Entry / exit / stop and rollback: Before launch each proposed role must be assigned to an actual person. Report owner checks by 10:00 UTC, ack by 12:00 UTC and manual escalation to operations lead thereafter. No automated messaging is added. Unknown publication: lock out rerun, inspect final manifest/hash under read-only check, document disposition. Stale lock: confirm no active process and inspect report before manual removal. Correction: archive old date directory with owner approval outside this job, preserve evidence then rerun. Retention deletion is a separate manual procedure.

Experiment: Owners approve rollout and residual risk explicitly; acceptedRisk remains false in this design.

10 Coding handoff

X1 - Build the bounded daily CSV report

Acceptance 1: CTX2/F1: implement a local deterministic CLI only; zero LLM/network calls, emails, remote updates, forecasts or conversation.

Acceptance 2: T1/C1: accept only allowed no-symlink paths, completed UTC date and bounded immutable file bytes; enforce exclusive date lock and exact run/config/input hashes.

Acceptance 3: T2/C2: enforce H1/H4 over complete files, exact IDs, unique headers/IDs, valid UTC timestamps and promised match for every delivered actual; fail entire run with exact reason/row and no final report.

Acceptance 4: T3/C3: all and only delivered actual rows inside H3 half-open day, strict integer-microsecond >24h, negative deltas false, sorted stable output; EV1/EV2/EV6/EV19 exact CSV bytes pass.

Acceptance 5: T4/C4/C5: committed report is immutable with COMPLETE manifest+matching CSV SHA256; identical rerun verified no-op, conflict never overwrite; uncertain publication readback before at most one retry; EV13-EV18 and EV23-EV27 pass.

Acceptance 6: T5/C6: proof-required exit 0; failure exit 2, cancellation 130; local owner/deadline/next action, stderr fallback and visible cleanup warnings; no remote notification.

Acceptance 7: FM1-FM9/EV1-EV32: implement future isolated harness and store actual results; no synthetic assertion is counted as executed before harness runs; release suite enumerates all EV1-EV29.

Acceptance 8: R4/R6: owners verify H1-H8 and EV32 evidence before consumption, execute shadow then assist-only runbook and manual rollback; no hidden schedule provisioning or permission expansion.

Acceptance 9: F5/EV29: persist only five output columns, hashes/counts and error row numbers; no raw-input copies/contents or extra customer fields in output/status/stderr.

Acceptance 10: ISSUE2/ISSUE3 repair: deterministic exact CTX4 manifest/config/run key JSON bindings; full stored manifest checks, distinct -a0/-a1 stage paths, and post-start cancellation always reconciled.

Build instructions: Build a Python 3 standard-library CLI with modules for C1 acquisition/config/locks, C2 strict CSV validation, C3 join/render, C4 staged immutable publication, C5 readback reconciliation and C6 local outcomes. Start at W1 and implement the exact enums/exclusive transitions through W9/W10/W11. Use strings for order IDs, aware UTC integer-microsecond timestamps and fixed decimal seconds; no float comparison. State ownership: C1 owns immutable snapshot/lock/run identity, C2 maps, C3 deterministic bytes/counts, C4 final write, C5 proof, C6 terminal status. Include stable config_digest/run_key canonical sorted compact JSON serialization and complete manifest shape from CTX4. Capability scope, no-symlink and trusted-host assumptions are F4/H6; no credentials or network adapters are required. Every expected error is explicit in C1-C6; an unexpected exception must block without inventing completion. Existing status failure cannot fabricate rollback of a proved final report. Implement in isolated /sandbox roots for EV1-EV29; materialize each complete fixture, compute hashes from its exact bytes, capture ordered mock calls and assert expected records and forbidden effects. Add EV30 component/property classes and fault injection on selected deployment filesystem. Store actual executed results separately; do not mistake design_io validation for a business harness. Verify H1-H8 source/schema/timing/permissions/retention/correction dependencies with proposed owners, then use EV32 release gate and R4/R6 runbook. Manual rollback disables local schedule and returns to manual join without deleting evidence. No email, messaging, remote update, forecasting, chat UI, order mutation, automatic overwrite or remote cloud provisioning. Persist stable error codes/row numbers only. Open questions are explicit launch dependencies, not permission to invent source guarantees. Use the exact CTX4 manifest and canonical digest fields. C4 stages at an attempt-specific -a0/-a1 path and signals after rename begins return UNKNOWN for C5 proof. Never reuse an orphan stage or treat matching filenames as a matching report.

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

A Assumptions and launch dependencies

H1 - LAUNCH BLOCKER

The proposed CSV contract is UTF-8 with optional BOM, comma delimiter and one header; promised columns are order_id,promised_delivery_utc and actual columns are order_id,actual_delivery_utc. Extra columns are ignored. IDs are case-sensitive ASCII [A-Za-z0-9][A-Za-z0-9_-]{0,63}, unique within each file; empty or padded IDs are errors.

Why: Exact schemas and an ID profile were not supplied; strict validation avoids accidental joins and spreadsheet formula execution.

Proposed owner: Operations data owner

Required validation: Inspect representative exports, including leading-zero IDs and unusual ID characters; approve this contract or version a replacement before launch.

H2 - LAUNCH BLOCKER

Operations supplies two complete snapshots from the same order namespace and consistent export cutoff after the requested day ends. CSVs have no authoritative dataset/as-of metadata; the job cannot independently prove their business freshness or shared provenance.

Why: A correct join can still be wrong when files come from different sources or omit recently recorded deliveries.

Proposed owner: Operations data owner

Required validation: Verify export procedure and late-arrival/correction timing; record source, cutoff and pair-selection evidence in the operating runbook before use.

H3 - LAUNCH BLOCKER

The daily report covers actual deliveries in the previous UTC calendar day by default, with an explicit --report-date override for backfills. It contains all delivered rows in that day, not only late rows; a boolean identifies lateness.

Why: Daily window and output inclusion were unspecified. One row per delivered order supports reconciliation and shows on-time results.

Proposed owner: Operations report owner

Required validation: Confirm daily window and all-deliveries output on representative manual examples before launch.

H4 - LAUNCH BLOCKER

Accepted timestamps use YYYY-MM-DDTHH:MM:SS[.ffffff]Z or +00:00, with 1-6 fractional digits, valid Gregorian dates and seconds 00-59. Empty actual means undelivered; non-UTC offsets, missing promised values, leap seconds and greater precision fail validation.

Why: ISO UTC is broader than a single parser profile; exact bounded precision gives deterministic arithmetic.

Proposed owner: Operations data owner

Required validation: Check real timestamp samples and establish this profile; unsupported valid ISO forms require an explicit parser-contract revision.

H5 - LAUNCH BLOCKER

Proposed operational budget is at most 100,000 rows and 50 MiB per file, 120 seconds per run, and a once-daily invocation at 08:00 UTC after exports are ready. Operations reviews local status by 10:00 UTC and acknowledges failures by 12:00 UTC; after that the operations lead manually escalates.

Why: Small, daily and operations ownership need concrete provisional limits and a local failure disposition.

Proposed owner: Operations report owner

Required validation: Measure actual sizes/runtime and confirm schedule, staffing, acknowledgement and escalation practice before launch; tune versioned limits if necessary.

H6 - LAUNCH BLOCKER

Run on one trusted local host/account with two allowlisted input roots, a dedicated output root and same-filesystem staging. Only the job may mutate its output root; no symlinks are permitted in resolved input/output paths. An operator grants these local capabilities in configuration.

Why: Local filesystem isolation and atomic-directory publication are proposed controls, not verified deployment facts.

Proposed owner: Implementation owner

Required validation: Exercise permissions, exclusive locks, fsync and directory rename/crash behavior on the chosen filesystem; confirm trusted-account boundary.

H7 - LAUNCH BLOCKER

Keep committed CSVs, manifests and local failure records for 30 days. Do not copy raw input files; persist hashes, counts and error row numbers without row contents. Delete expired files locally only through a separate operator-maintained procedure.

Why: No retention policy or personal-data classification was supplied; minimization and a provisional retention period need approval.

Proposed owner: Operations data owner

Required validation: Approve access, retention and order-ID sensitivity before launch; verify any stricter organizational requirement.

H8 - LAUNCH BLOCKER

A date with an existing valid completed report is immutable: identical input/config returns ALREADY_COMPLETE; changed input/config returns CONFLICT. Corrections require the owner to archive the old date directory outside this job and rerun; no replacement action is implemented.

Why: Silent overwrites can cause the operations team to act on inconsistent reports.

Proposed owner: Operations report owner

Required validation: Approve the manual correction/runbook rule and prove conflict behavior in a local staging environment.

Other unique recorded assumptions

models: Zero LLM components and zero model calls/tokens/prices: all leaves are deterministic code. No fine-tuning, prompt budget, LLM retry, provider account or model spend is applicable. Runtime/storage/operator costs are unmeasured and excluded; zero LLM usage is not a claim of zero operating cost.

B Review evidence and limits

Review method: self-review; round 3; supplied status pass.

Dimension	Score	Supplied evidence
D1	3/4	CTX1-CTX4/T0-T5 bound a daily join/export with exact exclusions and owned cancellation/failure.
D2	3/4	Every leaf maps to C/W/EV; the generated all-category FM□EV relation is reciprocal; H1-H8 distinguish assumptions from the unchanged brief.
D3	3/4	W1-W11/C4/C5: exhaustive outcomes, one proven-absent retry with distinct staging path; post-start cancellation always readback; exact final proof before completion.
D4	3/4	C1-C6 and CTX4: exact typed manifest/config/run-key canonical object fields, contracts and errors; full stored manifest equality and local human disposition.
D5	3/4	C2/C3 own dates/IDs; F4 local allowlists and C4 fresh checks; manifest proof and H7 minimized retention. H6 platform behavior remains unverified.
D6	3/4	H1-H8 mark source format, window, volumes, owners, policy and filesystem behavior as proposed launch dependencies; no vendor/current-fact claims.
D7	3/4	F1-F5 choose code/local filesystem over LLM/database/cloud and compare deployment alternatives on identical criteria without claiming integrations.
D8	4/4	Models.items is empty with an explicit zero-LLM budget; runtime/storage/human costs excluded as unmeasured.
D9	3/4	FM1-FM9 are case-specific with S/O/D and owners; local unknown commits, injection, conflicts, failure visibility and privacy have explicit tests.
D10	3/4	EV1-EV29 self-contained concrete fixtures; EV30 component/property layers, EV31 monitor and EV32 exact golden membership. Business harness is future work.; Actual independent fixture-label arithmetic audit inspected EV1/EV2/EV6/EV19/EV24/EV29 expected CSV rows: all passed; this is not execution of the future business harness.
D11	3/4	R1-R6 define denominator-based KPIs, source-baseline dependency, noncircular shadow/assist gates, continuous paired effort experiment and manual stop/restore.
D12	3/4	X1/CTX4/C1/C4/C5 propagate repairs; actual helper validation exited 0 with zero errors/warnings. Same-folder export round 1 succeeded; final export inspection is recorded separately in check-evidence.

ISSUE1 / major / resolved / export/X1

Actual helper validate and same-folder export both exited 0; round-1 evidence is saved. Semantic payload review then found two material implementation ambiguities below.

ISSUE2 / major / resolved / components/C4

CTX4/C1/C4 did not give one exact typed manifest object or exact config-digest key set, yet C4 delegated the manifest to CTX4. Different implementers could serialize different identity/config/count fields. Specify exact fields/canonical JSON and validate full stored binding before a no-op/completion. Repaired: CTX4 specifies exact manifest shape and canonical config/run-key bytes; C1/C4/C5/X1 propagate the binding.

ISSUE3 / major / resolved / workflow/W6

C4 reused one staging path across its allowed retry after proven absence, so an orphan first-attempt staging directory could block EV15 despite the designed retry. Also EV27 expected explicit reconciliation for a signal after rename start without an explicit C4 output rule. Give attempt-specific stage paths and a precise post-start cancellation outcome. Repaired: C4 uses -a0/-a1 staging paths, fresh retry staging, owned cleanup, and explicit UNKNOWN on post-start cancellation; C5/X1 agree.

AI self-review only, not stakeholder sign-off or independent review.

No business report is implemented, scheduled or executed; all EV1-EV32 business tests are specifications.

Real exports, source freshness, operating owners/limits, local filesystem guarantees and retention need H1-H8 evidence before launch.

Design helper checks structure/relationships and renders documents; it does not prove fixture truth, business correctness or production impact.

No source namespace/as-of metadata exists in the two CSV contract; H2 remains a real manual provenance/freshness prerequisite. Hashes alone cannot establish it.

Design-only tooling: this helper makes no provider, network or business API calls and executes no business assertions; optional host-assistant research is outside this statement.

Schema, reference, graph and ID-coverage checks do not establish semantic completeness, source truth, actual human authorization or production safety.

Review scores and evidence are supplied by the reviewing agent; the helper checks their shape and declared pass conditions, not their truth.

FMEA action/control text is required where specified; the helper does not verify ownership, release authority, regulatory compliance or risk acceptance.

Canonical design SHA256: c6ee5c5f4fd25e5636635c684580b3c8f087a9b026a8034b6011cda7ab758ff6